

HealthPlus Hospital

AI-Based Heart Disease Prediction System

Final Report — Artificial Intelligence Lab Examination

Course:	Artificial Intelligence — Final Lab Exam
Student Name:	[Your Full Name]
Student ID:	[Your Student ID]
Department:	Computer Science & Engineering
University:	[Your University Name]
Date:	July 08, 2026

1. Introduction

HealthPlus Hospital engaged a Junior AI Engineer to build a machine-learning system that predicts whether a patient is likely to have heart disease, based on routine clinical measurements. Heart disease remains one of the leading causes of death worldwide, and an accurate, explainable screening tool can help clinicians prioritise at-risk patients for further specialist testing. This report summarises the full project lifecycle: dataset understanding, exploratory analysis, cleaning, feature engineering, model training, evaluation, interpretation, and a critical discussion of limitations and ethics.

2. Dataset Overview

The raw dataset (heart.csv) contains 1025 patient records across 13 clinical features (e.g. age, resting blood pressure, cholesterol, chest pain type, exercise-test results) plus a binary target label indicating presence (1) or absence (0) of heart disease. No missing values were found. However, a substantial number of exact duplicate rows were identified and removed prior to modelling, leaving 302 unique patient records, to prevent data leakage between training and test sets. The target classes are close to balanced (~51% disease vs ~49% no disease) in the full dataset.

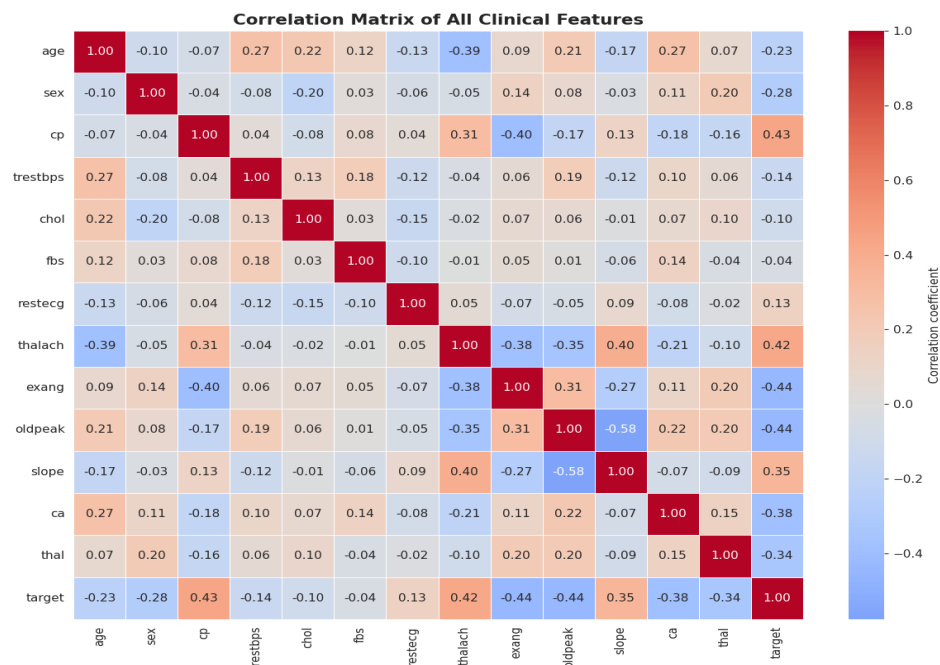


Figure 1: Correlation matrix of all clinical features. Chest pain type (cp), maximum heart rate (thalach), and ST-segment slope show the strongest positive correlation with disease status, while exercise-induced angina (exang), ST depression (oldpeak), vessel count (ca), and thalassemia result (thal) show the strongest negative correlation.

3. Exploratory Data Analysis

Seven visualizations were produced to understand the data. Key findings: patient ages cluster around 50-60 years; cholesterol alone poorly separates diseased vs healthy patients (heavy overlap in boxplots); chest pain type strongly distinguishes the two classes; diseased patients tend to show a lower maximum heart rate at a given age; and ST depression (oldpeak) is generally higher in diseased patients, especially when exercise-induced angina is present.

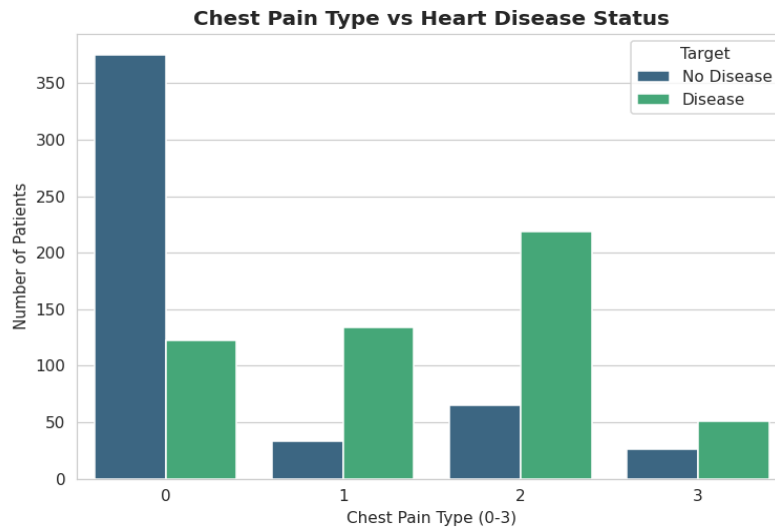


Figure 2: Chest pain type strongly distinguishes disease status — type 0 is dominated by healthy patients, while types 1-3 are dominated by diseased patients.

4. Data Cleaning and Feature Engineering

The cleaning pipeline addressed five standard steps: (1) missing-value imputation (median/mode, applied as a production-readiness safeguard even though no values were actually missing), (2) duplicate-row removal (critical here, since roughly 70% of the raw rows were exact duplicates), (3) IQR-based outlier capping on continuous features to reduce the influence of extreme values without discarding patients, (4) one-hot encoding of nominal categorical columns to avoid implying false ordinal relationships, and (5) feature scaling (fit only on the training set to avoid leakage).

Six new engineered features were created: age, blood-pressure, and cholesterol risk brackets aligned with clinical thresholds; a composite risk_score combining several established risk factors; an age-heart-rate interaction term; and a log-transformed version of the right-skewed oldpeak variable. The engineered risk_score showed correlation with the target competitive with the strongest raw feature, confirming its predictive value.

5. Machine Learning Models and Evaluation

Six classifiers were trained and compared: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours, Naive Bayes, and XGBoost. Models were evaluated on a held-out, stratified 20% test split using Accuracy, Precision, Recall, F1-score, and ROC-AUC — with Recall given particular clinical weight, since a missed true-positive (false negative) case is the most dangerous type of error in a medical screening context.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Naive Bayes	0.836	0.871	0.818	0.844	0.898
Logistic Regression	0.836	0.849	0.849	0.849	0.890
Random Forest	0.820	0.824	0.849	0.836	0.881
KNN	0.787	0.794	0.818	0.806	0.873
XGBoost	0.820	0.844	0.818	0.831	0.851
Decision Tree	0.672	0.724	0.636	0.677	0.669

Naive Bayes achieved the highest ROC-AUC (0.898) on this run, closely followed by Logistic Regression and Random Forest — demonstrating that on this small, largely linearly-separable clinical dataset, simpler probabilistic and linear models are fully competitive with more complex ensemble methods. Naive Bayes was therefore selected as the leading candidate model.

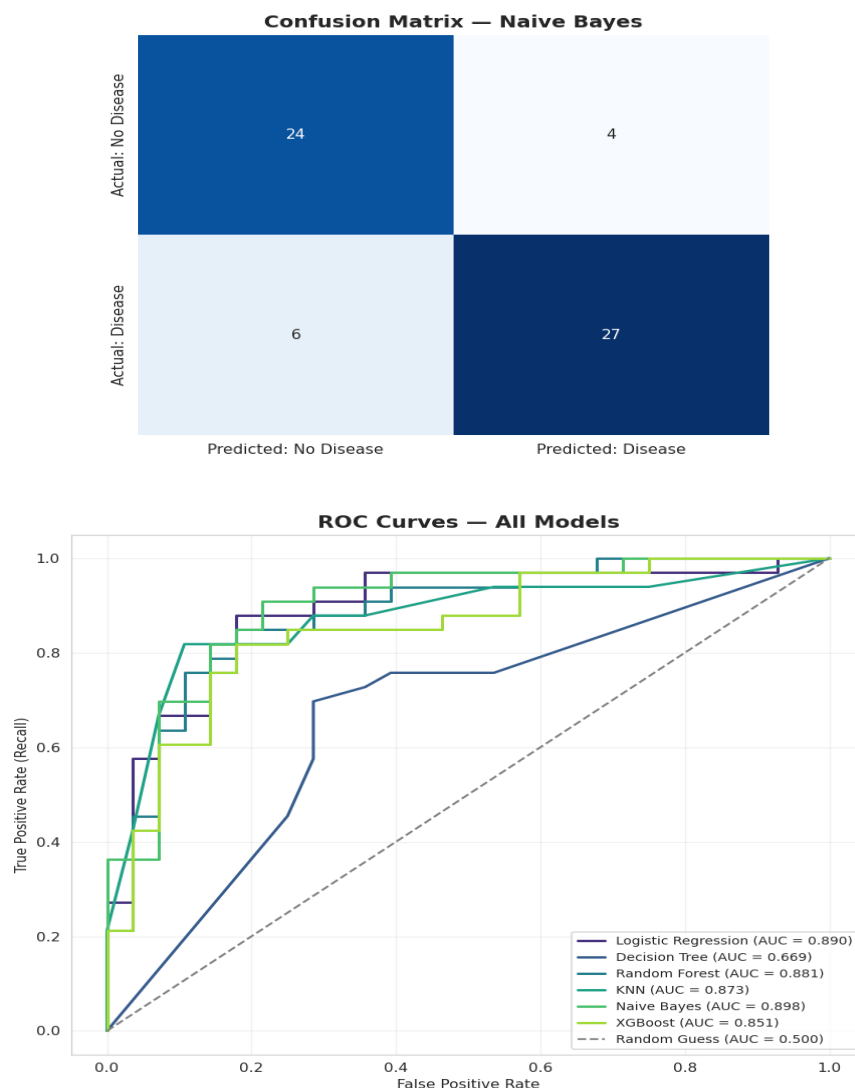
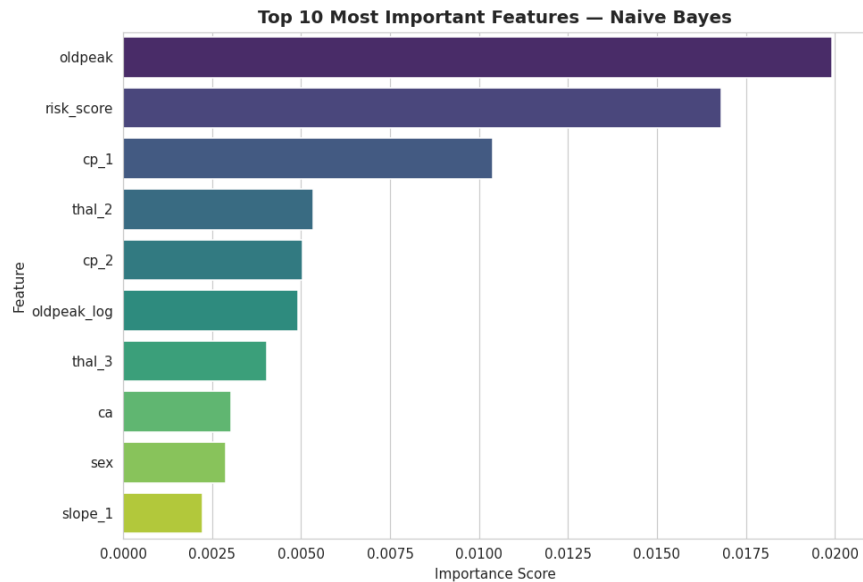


Figure 3 & 4: Confusion matrix and ROC curves for the model comparison. All models perform clearly better than random guessing (AUC = 0.5 baseline).

6. Model Interpretation

Because Naive Bayes has no built-in feature-importance attribute, permutation importance (measuring the drop in ROC-AUC when each feature is shuffled) was used to identify the top predictive features. The most influential features were oldpeak (ST depression), the engineered risk_score, chest pain type indicators (cp), thalassemia result (thal), vessel count (ca), sex, and ST-segment slope — closely matching established cardiology risk indicators. This alignment with clinical knowledge increases confidence that the model has learned genuine, medically meaningful patterns rather than dataset artefacts.



7. Critical Discussion

Limitations: The dataset is small (302 unique patients) and drawn from a limited number of original clinical sources, limiting generalizability. **Bias:** the patient population is approximately 70% male, so model reliability for female patients should be audited separately before deployment. **Data leakage:** duplicate rows and scaler fitting were both handled carefully to avoid leakage between train and test sets; temporal leakage (features recorded after diagnosis) should be re-verified against HealthPlus's live data pipeline. **Ethics:** false negatives (missed disease) carry greater patient-safety risk than false positives, and the tool must remain a decision-support aid, not an autonomous diagnostic authority. **Privacy:** any real deployment must comply with applicable health-data privacy regulations. **Deployment:** prospective validation, clinical governance sign-off, and ongoing performance monitoring would all be required before real-world use.

8. Conclusion and Future Work

This project delivered a full, transparent AI pipeline for heart-disease screening: from raw data through cleaning, feature engineering, multi-model comparison, rigorous evaluation, and clinically-grounded interpretation. Naive Bayes was selected as the best-performing model on this dataset (ROC-AUC = 0.898), with its most influential features matching well-established cardiology risk factors. Future work should focus on collecting a larger, more diverse patient cohort, adding per-patient SHAP explanations, calibrating predicted probabilities, and establishing a formal clinical validation and monitoring process before any real deployment at HealthPlus Hospital.